

LORENZO SANI

Cambridge, United Kingdom

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Summary

I am a Research Scientist at [Flower Labs](#) working on foundational models. I am also Ph.D. candidate in Computer Science at the University of Cambridge, working in the [Computer Laboratory](#) with the [CaMLSys](#) group led by Prof. Nicholas D. Lane. My research interests include **distributed and federated learning** on data centers and edge devices. I have experience optimizing the efficiency of **large-scale federated learning** settings, e.g., the **Photon** system and the **DEPT** method, whose insights can scale from edge devices to geo-distributed data centers.

Education

University of Cambridge

PhD in Computer Science

*Machine learning and system optimizations for federated pre-training of large language models.
Supervised by Prof. Nicholas D. Lane*

September 2022 – Ongoing

Cambridge, United Kingdom

Università di Bologna

Master of Science in Applied Physics

*Thesis titled "Unsupervised Clustering of MDS data using federated learning".
Supervised by Prof. Enrico Giampieri and Prof. Gastone Castellani*

December 2019 – March 2022

Bologna, Italy

Università di Bologna

Bachelor of Science in Physics

*Thesis titled "Correzioni relativistiche negli atomi idrogenoidi e struttura fine".
Supervised by Prof. Roberto Zucchini*

September 2016 – December 2019

Bologna, Italy

Selected Papers

Photon: Federated LLM Pre-Training

[Accepted MLSys'25](#)

Lorenzo Sani, Alex Iacob, Zeyu Cao, Royson Lee, Bill Marino, Yan Gao, Wanru Zhao, Dongqi Cai, Zexi Li, Xinchu Qiu, Nicholas D. Lane
This work presents the first complete system for federated end-to-end LLM training, leveraging cross-silo FL for global-scale training with minimal communication overheads.

DEPT: Decoupled Embeddings for Pre-training Language Models

[Accepted \(Oral\) ICLR'25](#)

Alex Iacob*, Lorenzo Sani*, Meghdad Kurmanji, William F. Shen, Xinchu Qiu, Dongqi Cai, Yan Gao, Nicholas D. Lane
DEPT decouples embeddings from the transformer body while simultaneously training the latter on multiple data sources without requiring a shared vocabulary.

The Red Queen Gödel Machine: Co-Evolving Agents and Their Evaluators

[arXiv](#)

Alex Iacob, Andrej Jovanović, William F. Shen, Daniel Burkhardt, Meghdad Kurmanji, Nurbek Tastan, Lorenzo Sani, Niccolò Alberto Elia Venanzi, Ambroise Odonnat, Zeyu Cao, Bill Marino, Xinchu Qiu, Nicholas D. Lane
A self-improving system in which agents and the evaluators that score them co-evolve, avoiding the reward hacking that fixed evaluators invite.

Flower: A Friendly Federated Learning Research Framework

[arXiv](#)

Daniel J. Beutel, Taner Topal, Akhil Mathur, Xinchu Qiu, Javier Fernandez-Marques, Yan Gao, Lorenzo Sani, Kwing Hei Li, Titouan Parcollet, Pedro Porto Buarque Gusmão, Nicholas D. Lane.
Flower is the most popular and friendly federated learning framework. It has more than 1000 citations.

Projects

SPRIN-D Composite Learning Challenge | *Python, PyTorch, Ray, gRPC, Flower*

February 2025

- The focus of this challenge is on developing solutions that enable **efficient model training on heterogeneous hardware**, from high-performance GPUs to CPUs of different types and manufacturers.

FlowerLLM/Photon | *Python, PyTorch, Ray, gRPC, Flower, SLURM*

May 2023

- Flower, in partnership with the [CaMLSys lab](#) at the University of Cambridge, has trained a **1.3 billion parameter LLM for the first time** using a novel formulation of federated learning methods. The first announcement can be found in [this blog post](#), while the research papers above summarize the most interesting results.

Flower Framework | *Python, PyTorch, Ray, JAX, TensorFlow, gRPC, Flower, SLURM*

May 2021

- [Flower](#) is a friendly federated AI framework with a unified approach to federated learning, analytics, and evaluation. I joined this project as an advisor and researcher in 2021, advising the main developers about potential new features and designs.

Work Experience

Flower Labs

January 2026 – Ongoing

Research Scientist

Cambridge, UK

- I work with the Model Team, developing methods and tools for foundational models training.

Flower Labs

May 2024 – September 2024; May 2025 – September 2025

Research Intern

Cambridge, UK

- I have developed methods and tools for efficient federated and large-scale pre-training of language models. I explored the extension of the Photon system to **ultra-efficient environments and to the development of MoE pre-training**.
- During the second internship at [Flower Labs](#), I have developed methods and tools for **efficient federated pre-training of language models**. I extended the Flower Framework to allow complete compatibility with the Photon system and contributed to designing the open-source framework's components.

- During my part-time contract with Oppent, I developed software for allowing the control and monitoring of the traffic control system managing a fleet of automated vehicles.

Other Papers

Flower Hub: A Reproducible Benchmarking Platform for Federated Learning in Simulation and Deployment

[arXiv](#)

Yan Gao, Mohammad Naseri, Javier Fernandez-Marques, Dimitris Stripelis, **Lorenzo Sani**, Davide Eynard, Fan Zhang, Hong Jia, Ting Dang, D. B. Emerson, Fatemeh Tavakoli, Ole Werger, Lars Wulfert, Petros Demetrakopoulos, Sofia Tsekeridou, InSeo Song, KangYoon Lee, Honghao Li, Lingjuan Lyu, John P Dickerson, Daniel Janes Beutel, Nicholas D. Lane
Flower Hub is a platform for publishing, discovering, and executing decentralized and federated applications, enabling reproducible benchmarking across simulation and deployment.

FoMoE: Breaking the Full-Replica Barrier with a Federation of MoEs

[arXiv](#)

Lorenzo Sani, Zeyu Cao, Meghdad Kurmanji, Alex Iacob, Andrej Jovanovic, Yan Gao, Wanru Zhao, Nicholas D. Lane
FoMoE breaks the full-replica barrier in distributed training by federating mixtures of experts, so no participant needs to hold the entire model.

Response-Conditioned Parallel-to-Sequential Orchestration for Multi-Agent Systems

[arXiv](#)

Nurbek Tastan, Alex Iacob, **Lorenzo Sani**, Meghdad Kurmanji, Nicholas D. Lane, Samuel Horvath, Karthik Nandakumar
A multi-agent orchestration scheme that adaptively transitions between parallel and sequential execution based on intermediate agent responses.

LoRDO: Distributed Low-Rank Optimization with Infrequent Communication

[Accepted ICML'26](#)

Andrej Jovanović, Alex Iacob, Mher Safaryan, Ionut-Vlad Modoranu, **Lorenzo Sani**, William F. Shen, Xinchu Qiu, Dan Alistarh, Nicholas D. Lane
LoRDO performs distributed low-rank optimization with infrequent communication, matching full-rank quality at a fraction of the synchronization cost.

MT-DAO: Multi-Timescale Distributed Adaptive Optimizers with Local Updates

[Accepted ICLR'26](#)

Alex Iacob, Andrej Jovanovic, Mher Safaryan, Meghdad Kurmanji, **Lorenzo Sani**, Samuel Horváth, William F. Shen, Xinchu Qiu, Nicholas D. Lane
MT-DAO combines optimizer moments tracked on multiple timescales with local updates, closing the gap to fully synchronous training under infrequent communication.

AbbIE: Autoregressive Block-Based Iterative Encoder for Efficient Sequence Modeling

[arXiv](#)

Preslav Aleksandrov, Meghdad Kurmanji, Fernando Garcia Redondo, David O'Shea, William Shen, Alex Iacob, **Lorenzo Sani**, Xinchu Qiu, Nicola Cancedda, Nicholas D. Lane
AbbIE is an autoregressive block-based iterative encoder that recursively refines representations, improving perplexity at a fixed parameter budget.

DES-LOC: Desynced Low Communication Adaptive Optimizers for Training Foundation Models

[Accepted ICLR'26](#)

Alex Iacob, **Lorenzo Sani**, Mher Safaryan, Paris Giampouras, Samuel Horváth, Andrej Jovanovic, Meghdad Kurmanji, Preslav Aleksandrov, William F. Shen, Xinchu Qiu, Nicholas D. Lane
DES-LOC desynchronizes the communication of optimizer states, cutting bandwidth for foundation-model training without sacrificing convergence.

LUNAR: LLM Unlearning via Neural Activation Redirection

[Accepted NeurIPS'25](#)

William F. Shen, Xinchu Qiu, Meghdad Kurmanji, Alex Iacob, **Lorenzo Sani**, Yihong Chen, Nicola Cancedda, Nicholas D. Lane
In this paper, we propose LUNAR, a novel unlearning methodology grounded in the Linear Representation Hypothesis. LUNAR operates by redirecting the representations of unlearned data to regions that trigger the model's inherent ability to express its inability to answer.

SparsityFed: Sparse Adaptive Federated Learning

[Accepted ICLR'25](#)

Adriano Guastella*, **Lorenzo Sani***, Alex Iacob, Alessio Mora, Paolo Bellavista, Nicholas D. Lane
SparsityFed is a practical federated sparse training method that critically addresses problems such as consensus over sparse masks among clients and robustness to nonIID data.

The Future of Large Language Model Pre-training is Federated [FM@FL NeurIPS '24 Workshop \(Oral\)](#)

Lorenzo Sani, Alex Iacob, Zeyu Cao, Bill Marino, Yan Gao, Tomas Paulik, Wanru Zhao, William F. Shen, Preslav Aleksandrov, Xinchu Qiu, Nicholas D. Lane
Initial technical report on my work related to federated pre-training of large language models.

Worldwide Federated Training of Language Models

[FM@FL NeurIPS '24 Workshop \(Oral\)](#)

Alex Iacob, **Lorenzo Sani**, Bill Marino, Preslav Aleksandrov, William F. Shen, Nicholas D. Lane
*We propose a Worldwide Federated Language Model Training (WorldLM) system based on federations of federations, where each federation has the autonomy to account for factors such as its industry, operating jurisdiction, or competitive environment. **Outstanding Paper Award.***

Fedanchor: Enhancing federated semi-supervised learning with label contrastive loss for unlabeled clients

[arXiv](#)

Xinchu Qiu, Yan Gao, **Lorenzo Sani**, Heng Pan, Wanru Zhao, Pedro Porto Buarque Gusmão, Mina Alibeigi, Alex Iacob, Nicholas D. Lane
We propose FedAnchor, an innovative Federated Semi-Supervised Learning method that introduces a unique double-head structure, called anchor head, paired with the classification head trained exclusively on labeled anchor data on the server.

Sheaf hypernetworks for personalized federated learning

[arXiv](#)

Bao Nguyen, **Lorenzo Sani**, Xinchu Qiu, Pietro Liò, Nicholas D. Lane
We propose a novel class of HNs, sheaf hypernetworks (SHNs), which combine cellular sheaf theory with HNs to improve parameter sharing for PFL.

Pollen: High-throughput simulation of federated learning via resource-aware client placement

[arXiv](#)

Lorenzo Sani*, Pedro Porto Buarque Gusmão*, Alex Iacob*, Zhao Wanru, Xinchu Qiu, Yan Gao, Javier Fernandez-Marques, and Nicholas D. Lane
This work proposes Pollen, a novel resource-aware system for speeding up federated learning simulations in highly heterogeneous resource environments.

D6.2 - Preliminary conclusions about Federated Learning applied to clinical data

[Zenodo](#)

Álvarez Federico, Zazo Santiago, Parras Juan, Almodóvar Alejandro, Alonso Patricia, Giampieri Enrico, Castellani Gastone, **Sani Lorenzo**, Rollo Cesare, Sanavia Tiziana, Krogh Anders, Prada-Luengo Íñigo, Kanterakis Alexandros, Sfakianakis Stelios, Cremonesi Francesco
This report comprises the first contributions from different partners of the GenoMed4All H2020 Project on Federated Learning (FL).

Teaching Experience

L361 - Federated Learning Theory and Practice.

January 2023 – March 2025

Teaching Assistant

Cambridge, UK

- Course ran for **three academic years** during Lent Term (from January to March): 2023, 2024, 2025
- **First-ever university-level academic course in Federated Learning.**
- We expanded students' machine learning expertise by examining the manifestation of concepts in decentralized settings, encompassing theoretical aspects like decentralized optimization and practical considerations like networking efficiency.
- Course on Federated Learning technologies taught by Prof. Nicholas D. Lane to PatII, PartIII, and MPhil students at the Department of Computer Science and Technology of the University of Cambridge.

L46 - Principles of Machine Learning Systems.

October 2023 – December 2023

Teaching Assistant

Cambridge, UK

- We covered the principles and methodologies of scalable and efficient machine learning systems, covering algorithms and system techniques for training models across diverse computing environments.
- Course on Federated Learning technologies taught by Prof. Nicholas D. Lane to PartIII and MPhil students at the Department of Computer Science and Technology of the University of Cambridge.

Technical Skills

Languages/ML Frameworks: Python, Java, C++, HTML/CSS, JavaScript, PyTorch, TensorFlow, Jax

Developer Tools: VS Code, Cursor, Android Studio, Various Cloud Computing Infrastructures (e.g., Lambda Labs, Fluidstack, Runpod)

Technologies/Frameworks: Linux, GitHub, Slurm

Leadership / Extracurricular

U.S. Santos 1948 a.s.d.

2021 - Ongoing

Council Board Member

Reggio Emilia, Italy

- I am currently serving as a Council Board Member for my hometown's local non-professional football club.
- I mostly focus on digital processes and the club's administration.
- U.S. Santos 1948 a.s.d. counts more than 500 associates, more than 50 coaches, and tens of team managers.
- This activity is carried on as unpaid volunteering.

U.S. Santos 1948 a.s.d.

2011 - 2022

Football Coach

Reggio Emilia, Italy

- I served as a football coach for up to three teams per season for about 11 years.
- I had the privilege to coach from kids to adults, achieving the most important results in the history of the football club by winning as a deputy coach the playoff tournament of the non-professional league we were competing in.